

Project: TechFlow System Architecture

Overview

This document outlines the technical architecture for the TechFlow platform, focusing on scalability, reliability, and modular design. The system follows a microservices approach to ensure independent deployability and fault tolerance.

Core Components

1. Frontend Layer

The user interface is built as a Single Page Application (SPA) providing a responsive and interactive experience across devices.

- Framework: React with TypeScript
- State Management: Redux Toolkit
- Styling: Tailwind CSS

2. API Gateway

Serves as the single entry point for all client requests, handling routing, authentication, and rate limiting.

- Technology: Kong API Gateway
- Security: JWT-based authentication

3. Microservices

Each service is scoped to a specific business domain.

Service Name	Primary Responsibility	Tech Stack
Auth Service	Identity management, MFA, and session handling	Node.js / Express
Data Engine	Processing high-volume analytical data	Go
Notification	Multi-channel alerts (Email, Push, SMS)	Python / FastAPI
User Profile	Managing user preferences and social graphs	Node.js

Infrastructure & DevOps

Cloud Infrastructure

The platform is hosted on AWS, utilizing a multi-region deployment strategy for high availability.

- **Compute:** Amazon EKS (Kubernetes)
- **Networking:** Amazon VPC with private subnets
- **Storage:** Amazon S3 for static assets

Database Strategy

Different data models require specific database engines to optimize performance.

Data Type	Database Choice	Reason
Relational	PostgreSQL	ACID compliance for financial/user data

Data Type	Database Choice	Reason
Cache	Redis	Low-latency session and frequent query storage
Search	Elasticsearch	Complex full-text search capabilities

CI/CD Pipeline

Continuous integration and deployment are handled via GitHub Actions.

1. **Build:** Containerization using Docker.
2. **Test:** Automated unit, integration, and E2E tests.
3. **Deploy:** Blue/Green deployment strategy on Kubernetes clusters.

Monitoring & Observability

- **Logging:** ELK Stack (Elasticsearch, Logstash, Kibana)
- **Metrics:** Prometheus & Grafana
- **Tracing:** Jaeger for distributed request tracing

Security Compliance

The system is designed with a "Security-First" mindset, ensuring all data at rest and in transit is encrypted using AES-256 and TLS 1.3 respectively. Regular penetration testing and vulnerability scanning are integrated into the development lifecycle.